

VARUN NAUDIYAL

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PROFILE

I am a Master's student who works in the field of **Multi-Messenger Astrophysics**. My interests lie in using **machine learning and numerical simulations to explore the physics of Neutron Stars** and their evolution in various environments. Alongside this, I am also interested in understanding the **evolution and formation of stellar binary systems** and how their interactions can lead to the formation of compact binaries, whose mergers we ultimately observe through gravitational waves and electromagnetic counterparts such as kilonovae.

PROJECTS

Physics-Informed Neural Networks for Neutron Star EOS Inference (Ongoing) Aug 2026 – Present
Supervised by Dr. Ritam Mallick, IISER-B *Master's Thesis*

- Extending coupled PINNs for EOS inference to include tidal deformability alongside the TOV equations, enforcing EOS consistency with both.
- Building a framework to handle first-order phase transitions and the EOS discontinuities they introduce.
- Adding Bayesian inference to quantify uncertainty in the inferred EOS.

Rotational Energy Budget of Supramassive Neutron Star Merger Remnants Jun 2026 – Jul 2026
Supervised by Dr. Ritam Mallick, IISER-B *Summer Project*

- Built a Python pipeline around the RNS code to construct constant-baryonic-mass sequences of supramassive NSs, from the mass-shedding limit to the turning point, for any EOS.
- Computed the extractable rotational energy along these sequences to test the extent to which a kilonova could be “super-charged” by it.
- Compared against the GW170817 kilonova energy, implementing a consistency test for EOS which would be capable of reproducing the energies observed.
- Validated the result that a Supramassive Remnant would need to be near or at the mass-shedding limit.

Nuclear EOS: RMF/QHD Implementation for Neutron Stars (Ongoing) Aug 2026 – Present

- Implementing a beta-equilibrated RMF EOS with the baryon octet coupled to σ , ω and ρ mesons, with electrons and muons enforcing charge neutrality; self-consistent mean-field solver in progress.
- Extending to hyperon-rich and hybrid (hadron–quark) EOS to study effects on maximum mass and compactness.

Exploring Neutron Stars in R^2 Gravity Jun 2025 – Jul 2025
Supervised by Mr. Asim Kumar Saha and Dr. Ritam Mallick, IISER-B *Summer Project*

- Derived the Starobinsky R^2 field equations from the action, by hand and with a generalised Mathematica pipeline.
- Extended the TOV equations to $f(R)$ gravity and integrated them with a shooting method for a polytropic EOS to map mass–radius curves.
- Diagnosed the scalaron fine-tuning problem, i.e, exterior solutions are exponentially sensitive to the central Ricci scalar.

Numerical Relativity in 1 Dimension Aug 2025 – Nov 2025
Supervised by Mr. Hrishikesh Borthakur and Dr. Ritam Mallick, IISER-B *Semester Project*

- Studied 3+1 (ADM) numerical relativity and derived Oppenheimer–Snyder collapse in pressureless-dust, radiation-pressure and slow-rotation approximations.
- Evolved a spherical polytrope ($K = 100$, $\Gamma = 2$) with GR1D; verified stability by matching central-density oscillation modes to theory.

TECHNICAL SKILLS

Languages	Python, Fortran90, Mathematica, L ^A T _E X
Python Libraries	NumPy, SciPy, Matplotlib, Pandas
Codes	RNS (Rotating Neutron Stars), GR1D (GR 1D Hydrodynamics)
Numerical Methods	ODE integration (shooting, Runge–Kutta), root-finding, finite differencing

RELEVANT COURSEWORK

(PHY616) General Relativity	(PHY630) Cosmology I
(PHY633) Introduction to Astrophysics	(PHY634) Advanced Statistical Mechanics
(PHY615) Quantum Field Theory I*	

*ongoing

POSITIONS OF RESPONSIBILITY

Core Committee Member <i>IISER-Bhopal Astronomy Club (IBAC)</i>	Aug 2024 – Mar 2025
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- Contributed to the annual summer magazine, gave a public lecture on star formation, and ran public telescope sessions with Dobsonian reflectors.

Core Committee Member, Design & Content <i>Armageddon — Annual Tech-Fest, IISER Bhopal</i>	Aug 2024 – Nov 2024
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- Led the Design and Content team: posters, banners, the official fest logo, and scripts for event hosts.

Peer Counselor <i>IISER-Bhopal Student Wellness & Counselling Team</i>	Aug 2023 – Mar 2024
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- Provided confidential peer support to students and helped organise outreach to reduce stigma around mental health.